Paper II: best verified results available so far

Provisional evidence snapshot

NOT MANUSCRIPT-PROMOTED. This is a durable lower-bound snapshot of completed, exact-scored evidence. Live lanes remain unfinished; no missing result is interpreted as a failure or a zero.

What is already secured

Four full-horizon Hubbard–Holstein matched-accuracy pairs reach $t = 20$. Across those pairs, mean infidelity is 6.37×10^{-6}–1.12×10^{-5} for APM/exchange and 6.00×10^{-7}–2.08×10^{-6} for AVQDS. A separate completed AVQDS arm reaches 1.174×10^{-7}, but has no qualified APM mate.

Best individual $t = 20$ mean infidelity	1.174×10^{-7}, AVQDS, strong/fast parameter motion; **unpaired**
Best within a qualified matched pair	6.004×10^{-7}, AVQDS, fast/weak parameter-motion pair
Best observed $t = 20$ APM mean infidelity	6.370×10^{-6}, weak/slow parameter-motion pair
Matched-pair coverage	4 qualified pairs / 8 method arms, all reproduced on page 2
Observed logical-cost pattern	Within these four retained pairs, APM's recorded graph tuple and N_θ are componentwise smaller than AVQDS's

Interpretation boundary. These are observed, exact-referenced trajectories under the recorded response-normalized matched-accuracy contract. They establish an accuracy floor and a reproducible resource trend for the completed pairs; they do not establish universal method superiority or completion of the 39-cell prospective matrix.

Evidence semantics

- Accuracy quantities are physical errors against exact references; exact evolution is reporting-only.
- The resource tuple is $(\widehat{C}_{2q}, \widehat{C}_D, \widehat{C}_{1q}, N_\theta)$, a routing-independent logical/graph proxy.
- Qiskit compilation is excluded from this prospective campaign; implementation-parity evidence is therefore **OUT OF SCOPE**, not missing success evidence.

First-page evidence manifest

Evidence block	Rows	Audit / state	Scientific status
Canonical HH staged ledger	345	**PASS**	4 qualified $t = 20$ matched pairs; 15 qualified and 21 pending method arms overall
Prospective HH	1 scored	**BOUND**	AVQDS $t = 5$ individually qualified; comparison/cost claim pending APM mate
SIAM $N_b = 4$ pilot	2 scored	**BOUND**	numerical candidates only; no individually qualified matched pair yet
Live prospective lanes	3 states	**OPEN**	Hubbard RUNNING; HH APM PREFLIGHT_ONLY; SIAM r07 SUBMITTED

Canonical HH cluster set (31):

9697391, 9698201, 9698247, 9698264, 9701941, 9703334, 9703747, 9703851, 9705193, 9707629, 9708157, 9708462, 9708981, 9709593, 9709600, 9709707, 9709708, 9709713, 9709723, 9709731, 9709741, 9709744, 9709745, 9709750, 9709763, 9709772, 9709886, 9709947, 9709982, 9712867, 9713112.

Current prospective identities:

HH completed: `9715119.0`. Hubbard running at the last combined scheduler capture: `9714499.0`, `9714501.0`. SIAM scored: `9714484.0`, `9715147.0`; r07 submitted exactly once as `9715270` after that capture.

Durable sources:

Canonical report PDF SHA-256: `e0fdcef1aec522d4d3bfd54e9563842e8847f4608975a389946a60b010644929.`
Canonical report JSON SHA-256: `6461dc8255fb68276932bd2c7856aa43d856c0300d3f199b2e5d95305f5c8da9.`
Latest combined capture SHA-256: `64deab5ede574db7f29188913ea2fb7ed938669d068351cd06a96f3875a2476e.`

Full file hashes, source paths, claim bindings, and caveats are retained in the adjacent source-map and manifest JSON files. The canonical 14-page campaign report and one-page SIAM r06 report are appended unchanged after this three-page cover.

Completed full-horizon HH evidence: four matched $t = 20$ pairs

The table reproduces the eight method arms identified by the canonical report as four qualified response-normalized matched-accuracy pairs. Errors are time-mean absolute errors against the exact reference. $\mathcal{B}$ is the recorded aggregate normalized accuracy burden; it is an accuracy quantity, not a hardware cost.

Regime	Pair definition	Method	Mean infidelity	Mean $\lvert\Delta E\rvert$	Mean $\lvert\Delta D\rvert$	Mean $\lvert\Delta C\rvert$	$\mathcal{B}$	Logical tuple
weak / slow	parameter motion	AVQDS	2.084×10^{-6}	1.471×10^{-4}	1.362×10^{-4}	7.953×10^{-4}	1.421	$(5214, 5476, 9281, 945)$
weak / slow	parameter motion	APM/exch.	6.370×10^{-6}	8.593×10^{-5}	2.157×10^{-4}	4.281×10^{-4}	1.233	$(172, 254, 365, 53)$
weak / slow	state + parameter	AVQDS	1.200×10^{-6}	1.040×10^{-4}	9.707×10^{-5}	5.793×10^{-4}	1.009	$(5130, 5390, 9048, 926)$
weak / slow	state + parameter	APM/exch.	6.395×10^{-6}	1.040×10^{-4}	1.903×10^{-4}	3.892×10^{-4}	1.232	$(178, 248, 395, 53)$
weak / slow	state motion	AVQDS	1.603×10^{-6}	1.506×10^{-4}	1.447×10^{-4}	7.919×10^{-4}	1.461	$(1908, 2074, 3642, 370)$
weak / slow	state motion	APM/exch.	7.360×10^{-6}	1.328×10^{-4}	1.228×10^{-4}	8.193×10^{-4}	1.303	$(182, 256, 388, 54)$
fast / weak	parameter motion	AVQDS	6.004×10^{-7}	4.242×10^{-4}	1.182×10^{-4}	3.856×10^{-4}	0.062	$(170, 238, 371, 61)$
fast / weak	parameter motion	APM/exch.	1.121×10^{-5}	5.605×10^{-4}	1.914×10^{-4}	4.872×10^{-4}	0.107	$(96, 144, 246, 40)$

Accuracy floor

- The completed matched-pair evidence is already in the 10^{-6}–10^{-5} mean-infidelity range; the best completed individual arm reaches 1.174×10^{-7}.
- The corresponding mean absolute observable errors shown here are in the 10^{-5}–10^{-4} range for energy/doublon and the 10^{-4} range for the displayed charge-imbalance response.
- The best mean infidelity is not used as a stand-alone winner: matched response burden and the retained observable caps remain part of qualification.

Resource observation

- For every completed pair above, APM/exchange records fewer logical one- and two-qubit graph operations, lower logical depth proxy, and fewer terminal independent parameters.
- This is a routing-independent proxy comparison. No Qiskit compilation, backend routing, shot cost, or hardware-noise claim is made here.
- Final manuscript attribution remains contingent on the prospective fixed-code matched-pair campaign and its pre-submit audit.

Best individual accuracy record (not a cost comparison): strong/fast AVQDS parameter-motion, cluster `9698264`, mean infidelity 1.1744525×10^{-7}, $N_\theta = 62$, graph tuple $(182, 256, 386)$. Its run SHA-256 is `e00618453d30fad48854eae7a462ab229bfa528f808efea97dbbbed7056ca540`. It is deliberately excluded from the four matched rows above because no qualified APM mate exists.

Source: canonical 345-record Paper-II staged campaign report. The complete report, including every evaluated row, safeguards, pending arms, definitions, and plots, begins on page 4 of this bundle.

Prospective evidence and the live boundary

New exact-scored trajectories

Benchmark	Method	Horizon	Cluster	Mean infidelity	$\mathcal{B}$	Status
HH $N_{ph} = 1$, crossover/midres	AVQDS, cut 10^{-2}	$t = 5$	9715119	1.658×10^{-5}	0.1119	**QUALIFIED**; APM mate pending, so no cost comparison
SIAM $N_b = 4$	AVQDS, r02 cut 10^{-3}	$t = 1$	9714484	2.602×10^{-5}	0.4380	CANDIDATE; broad guard passes, individual target fails
SIAM $N_b = 4$	APM/exchange, r06 cut 10^{-5}	$t = 1$	9715147	2.237×10^{-6}	1.5954	REJECTED PAIR; broad response guard fails

Verified detail

HH $t = 5$: mean $|\Delta E| = 3.315 \times 10^{-3}$, mean doublon error 9.671×10^{-4}, mean maximum site-occupation error 1.894×10^{-3}, and logical tuple $(270, 486, 617, 221)$. Exact source, seed, and source epoch are digest-bound; Qiskit compilation was not performed.

SIAM gate detail: AVQDS's worst primitive response-normalized mean error is 0.07938; APM r06's is 0.31897 in the small-response C_3 channel. Their mean $|\Delta E|$ values are 5.145×10^{-4} and 4.036×10^{-4}. The pair is accuracy-confounded and rejected; r07 is the pure planner's next APM rung.

Security statement. If every open lane stopped today, this bundle would still retain four complete HH $t = 20$ matched pairs, the unpaired 1.174×10^{-7} HH accuracy record, and one independently qualified prospective HH $t = 5$ trajectory, all with exact-reference accuracy and source hashes.

What this PDF does not claim

- It is not the final Paper-II table or 39-cell manuscript result set.
- It declares no SIAM winner and infers no threshold monotonicity.
- It reports no Qiskit-compiled, routed, shot, or hardware cost.
- It does not reopen the superseded adaptive FS-radius experiment.

Open work is explicit

Lane	Typed state	Consequence
HH APM mate	**PREFLIGHT**	dry-run only; no result
Hubbard L4	**RUNNING**	9714499/9714501; no score yet
SIAM r07	**SUBMITTED**	9715270; result pending
Full matrix	**INCOMPLETE**	final promotion not claimed

Audit anchors

- canonical campaign: **PASS**, 345 immutable rows;
- prospective HH: exact/seed/source identities joined and score digest verified;
- SIAM: exact source epoch and 101-point grid digest bound;
- latest combined scheduler capture: 2026-08-30 03:44:09 UTC;
- source evidence revision: `d5590449`.

Reader map. Pages 4–17 are the unchanged 14-page canonical campaign report. Page 18 is the unchanged SIAM r06 results sheet. The adjacent JSON sidecars bind every source and every headline claim to a SHA-256 digest.